RCC Detailed

Batch Copy

Why is RCC used instead of PCC everywhere?

Why is PSC not used everywhere?

What is the full form of PCC RCC & PSC

What are the strengths & weaknesses of the three

What concept does prestress use?

Why was only steel used for Reinforcing cement?

What is the value of α_{conc}^{2}

What is thermal expansion coefficient also called?

What analogy can be given for concrete & steel compatability?

What code to use for PCC structres?

What is the name of the code?

Or full form of the code!

12 March 24 RCC Class 1

PCC, RCC and Prestress design

PCC: plan cement concrete
- Good in compression only

RCC: Reinforcement cement concrete
- Good in compression as well as in tension.

PSC: Prestress concrete
advance technique uses pretensioning and post tensioning concept.

Ans: The thermal expansion coefficient of steel as well as concrete is almost same ie $\alpha_{conc} : \alpha_{steel} = 12 \times 10^{-6}/°C$

Thermal expansion coefficient is also called thermal contraction coefficient.

IS 456 : 2000 :
- Indian standard Plain & reinforced concrete code of practice.

Density or unit weight

(i) Cement : 1440 kg/m³
(ii) Aggregate : 1550 kg/m³
(iii) Sand : 1680 kg/m³
(iv) Brick masonry : 1920 kg/m³
(v) PCC : 2400 kg/m³
(vi) RCC : 2500 kg/m³
(vii) Steel : 7850 kg/m³

Concrete:

- Concrete is mixture of cement + sand + aggregate + water.
- Concrete is a heterogeneous material.
- Quality control is poor.
  - because it is made onsite.
- More factor of safety is taken.

Steel

Manufactured in factory hence quality control is good & proper

Steel is a homogeneous material

less factor of safety

Design methodology

WSM

Working stress mthd

- Deterministic approach
- factor of safety is used.
- Based on maximum principle stress theory (Rankines theory)

LSM

Limit state method (strain

- Probabalistic approach
- Partial safety factor for material as well as load is considered.
- Based on maximum principle strain theory (Venants theory)

Factors of safety definiton.

Factor of safety is the factor by which yield stress of material of a member divided by permissible stress in material. $F.o.S = \frac{\text{Yield stress}}{\text{Permissible stress}}$

(1) FOS : $\frac{\text{Yield stress}}{\text{Working stress}}$ or $\frac{\text{(V)S}}{\text{Permissible stress}}$

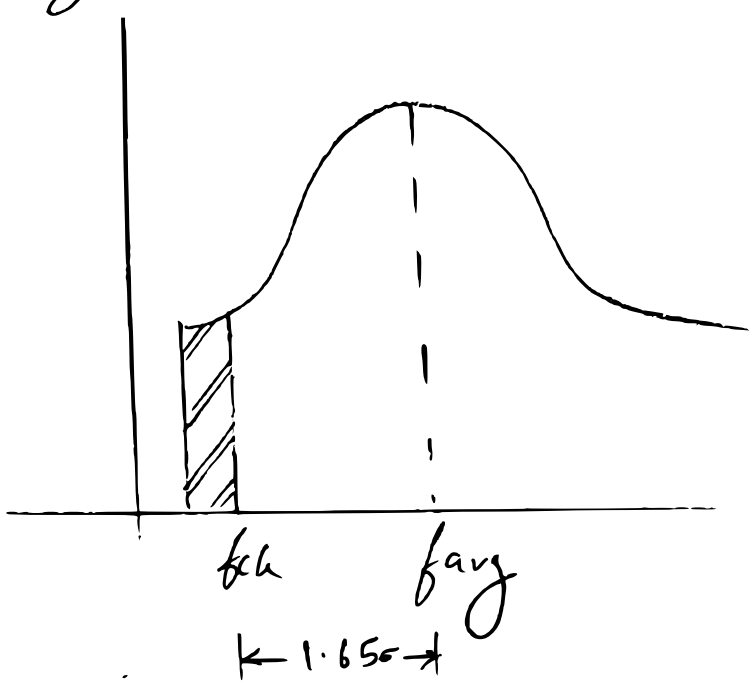
FYP = Five year plan.

FOS	WSM	LSM
steel	1.8	1.15
concrete	3.0	1.5

Limit state takes or considers lower or lesser factor of safety than WSM.

Characteristic strength of concrete (f_{ck})

Characteristic compressive strength of conc. The strength below which not more than 5% test results are expected to fail or fall is called characteristic compressive strength of concrete.



f_{avg} or f_m

$= f_{ck} + 1.65\sigma$

f_{avg} : avg str

f_m : means to

σ = standard deviation

Characteristic load: it is defined as the load which has high probability to remain same, upto 95%.

Compressive strength test of concrete (cube test)

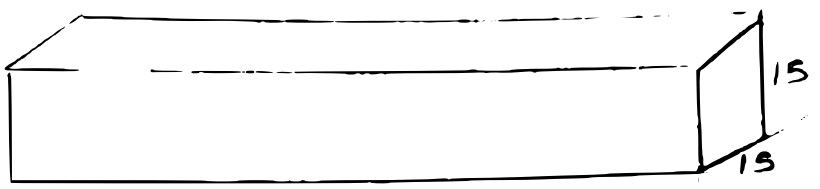
Size of cube used to test compressive strength 15 x 15 x 15 cm^3

in UTM.

flexure or bending or rupture strength of concrete.

Beam test 15cm x 15cm x 70cm.

2 point load test.



flexural/tensile strength.

$f_{cr} = 0.7 \sqrt{f_{ck}}$

(M)(20) ~> f_{ck}

~ mix.

for M25 $f_{ck} = 0.7 \times \sqrt{25}$

$= 3.5 \ N/mm^2$

Tensile strength of concrete test

1. Direct test.
   - Briquette test.
2. Indirect test.
   - Cylinder Splitting test.

Concrete mix proportioning

The calculation of proportion of constituents like cement, fine aggregate, coarse aggregate, water to achieve desired degree of workability of fresh concrete & desired strength, durability, surface finish of hardened concrete, mix proportioning is done.

Types of concrete mix

(1) Design mix concrete: It is based on weight batching & it is always preferable for concrete grade more than M20.

(2) Nominal mix concrete: It is based on certain thumb rule of mix constituents used for concrete grades upto M20. (including M20)

M25	1	1	2
M20	1	1.5	3
M15	1	2	4
M10	1	3	6
M7.5	1	4	8
M5	1	5	10

Nominal mix is based on volume batching.

Weight batching is always preferable to volume batching because in volume batching due to moisture volume increases specially in case of sand (sand bulking).

Batching refers to — controlling the quantity of each material into each batch. or the measurement of material for making concrete is called batching. It is essential to ensure uniformity of proportion & aggregate grading in successive batches

Grades of concrete Is 456 pg 16.

Table number 2

Group	Grades designated			
(1) Ordinary concrete	M10	M15	M20	
(2) Standard concrete	M25	M30	M35	M40
	M45	M50	M55	M60
(3) High strength concrete	M65	M70	M75	
	M80	M85	M90	M95
	M100			

Note: Cube test is done in cube size 150x150x150 mm for 28 days

Short term modulus of elasticity of concrete $E_c = 5000 \sqrt{f_{ck}}$

Variation in $E_c = \pm 20\%$.

Long term modulus of elasticity of concrete

$$E_{c\theta} = \frac{E_c}{1+\theta}$$

θ is creep coefficient

Age of loading	θ or creep coeff
7 days	2.2
28 days	1.6
$\geq$ 1 year	1.1

The ultimate creep strain does not include elastic strain.

$$\theta = \left(\frac{\text{ultimate creep strain}}{\text{elastic strain at the age of loading}} \right)$$

Modulus of elasticity of steel

$= 2 \times 10^5$ Mpa

2×10^6 kg/cm^2

or 200 GPa.

Tolerance limit

Variation allowed in modulus of elasticity for Concrete $\pm$ 20%.

$E_c \longrightarrow \pm 20\%$.

Workability of concrete: It is defined as the ease with which concrete can be handled, transported, placed, compacted.

Very imp. Placing condition	Degree of Workability	
(1) Blinding concrete, (i) Shallow section, (f) Pavement using paver	very low workability	Compaction factor 0.75 to 0.80.
(2) Mass concreting (Dam) lightly reinforced section, hand placed pavement, canal lining, strip footing	Low workability	25 - 75 mm (slump)
(3) (a) Heavily rf section in slab, beam, wall, column. (6) Slipform, pumped concrete	Medium workability	Slump cone test @ 50 to 100 @ 75 to 100
(4) trench fill	High	100 - 150
(5) In situ piling or tremie concreting	Very high	determined by flow table test

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13 March 2024. - class 2

Syllabus

1. Introduction
2. Design consideration for struct. member
   - A Beam — bending, shear, torsion, bond
   - B column
   - C Slab
   - D footing
3. Masonry works
4. Prestress design PSC

limit state design - Beam

limit state method

- Is of strength or collapse or safety
  1) flexure or Bending or Rupture
  2) Shear
  3) Torsion or twising
  4) Compression
- Is of serviceability
  1) Vibration
  2) Deflection
  3) Corrosion
  4) Cracking

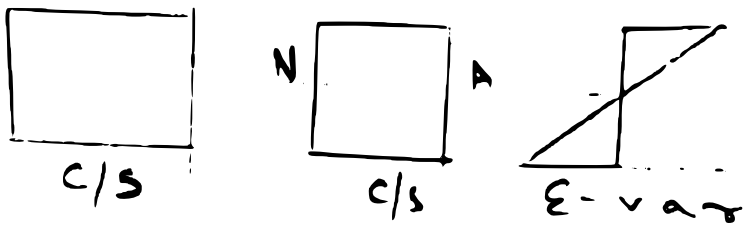
Note Durability is defined in IS456 (pg 17) as — factors affecting durability

1) The environment
2) The cover to embedded steel
3) Type & quality of constituent mat:
4) The cement content and w/c ratio of the concrete
5) Workmanship to obtain full compaction & efficient curing.
6) The shape and size of member.

limit state design - beam - bending

Assumptions

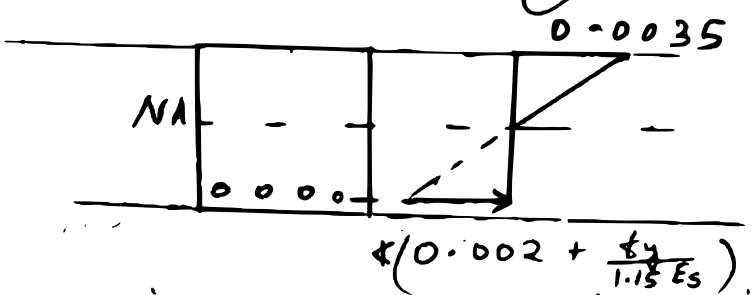
1. Plane section before bending remains plane after bending. ie strain distribution is linear or strain varies linearly with depth.



$\varepsilon \propto y^2$ or Strain α extreme fiber distance fm NA

Note: This assumption is valid for all design methods such as WSM LSM Prestress, ULM (ultimate load mthd)

2) The maximum strain in concrete at outermost compression fiber is taken as 0.0035. in bending or flexure.



3) The maximum strain in tension reinforcement at failure shall not be less than $0.002 + \frac{f_y}{1.15\,E_s}$ or ε_y

f_y : Characteristic strength / yield stress of steel

E_s : modulus of elasticity of steel

n less $\left(0.002 + \frac{f_y}{1.15E_s}\right)$

proof stress is taken as 0.2% proof strain

$\frac{0.2}{100} : \frac{2}{1000} = 0.002$

strain = stress/E

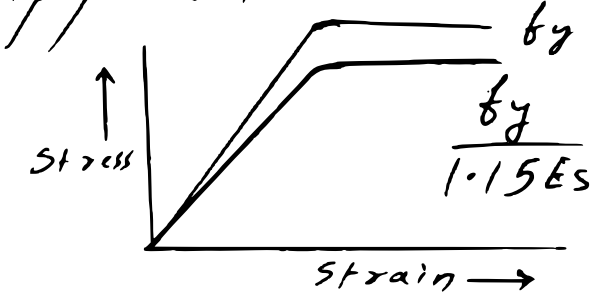
$\varepsilon = \frac{\sigma}{E} = \frac{f_y/1.15}{E}$

$= \frac{f_y}{1.15E_s}$

4. The tensile strength of concrete is ignored

5. The stresses in the reinforcement are derived from representative stress strain curve for the type of steel used.

for design purpose partial safety factor applied is 1.15.

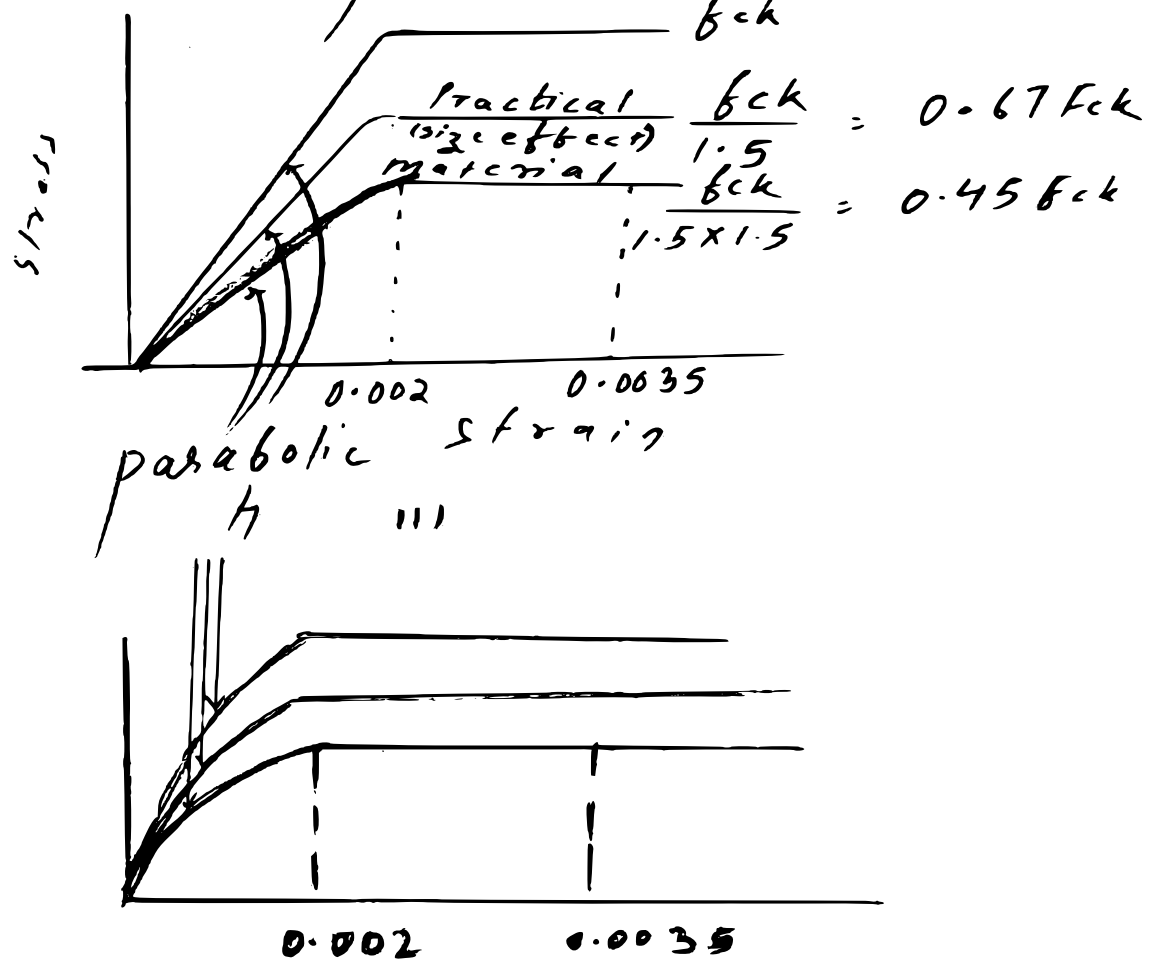


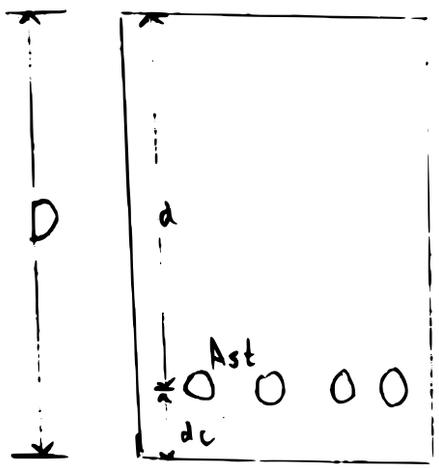
6. The relationship between compressive stress distribution in concrete and the strain in concrete (stress strain curve for concrete) may be assumed to be rectangular

trapezoidal or parabolic or any other shape which results in prediction of strength with the result of test.

For design purpose an acceptable Stress strain diagram for concrete is assumed parabolic.

The partial safety factor 1.5 shall be applied for characteristic strength and practical consideration.

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d
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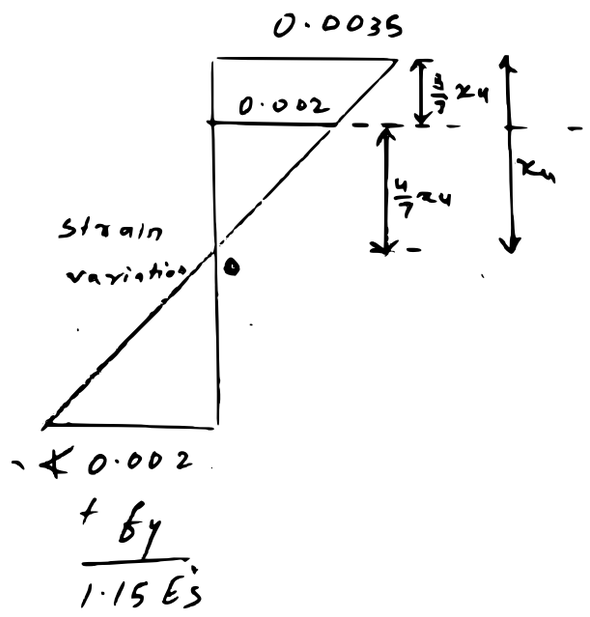
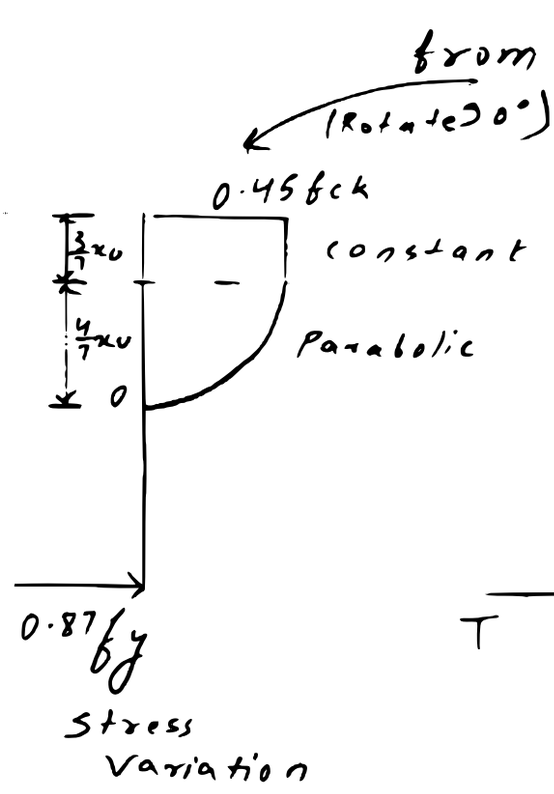
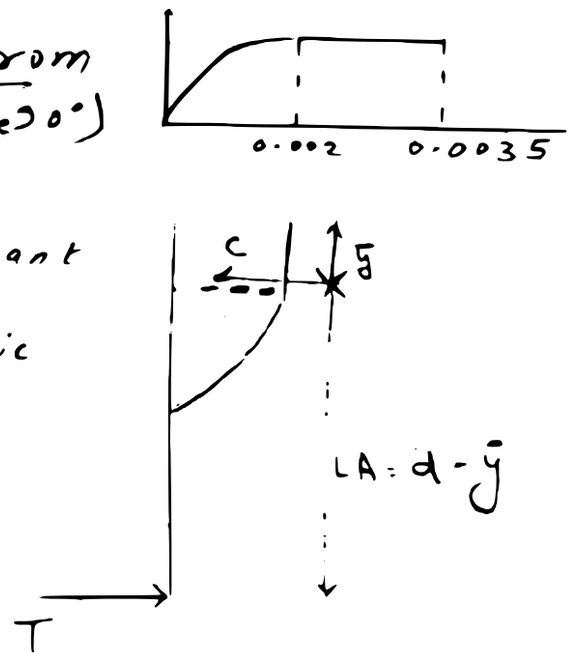

0.0035
0.002
$\frac{3}{7}x_u$
$\frac{4}{7}x_u$
x_u
Strain variation
0.002
$\frac{f_y}{1.15E_s}$


from
(Rotated 90°)
0.002
0.0035
0.45fck
$\frac{3}{7}x_u$
$\frac{4}{7}x_u$
0
constant
Parabolic
0.87fy
Stress Variation


C
T
LA = d - $\bar{y}$

ε_s (mess) $0.002 + \frac{f_y}{1.15 E_s}$ or $0.002 + \frac{0.87 f_y}{E_s}$

for mild steel / Fe 250.

$f_y = 250$

$\varepsilon_s = 0.002 + \frac{0.87 \times 250}{2 \times 10^5}$

$\Rightarrow \boxed{\varepsilon_s = 0.00308}$

for HYSD bar Fe 415

$f = 415$

$\varepsilon_s = 0.002 + \frac{0.87 \times 415}{2 \times 105}$

$\boxed{\varepsilon_s = 0.00380}$.

for Fe 500 $\varepsilon_s = \boxed{0.004 \frac{17}{..}}$

Fe 550 $\varepsilon_s = \boxed{0.00439}$

Analysis and design of beam or bending member.

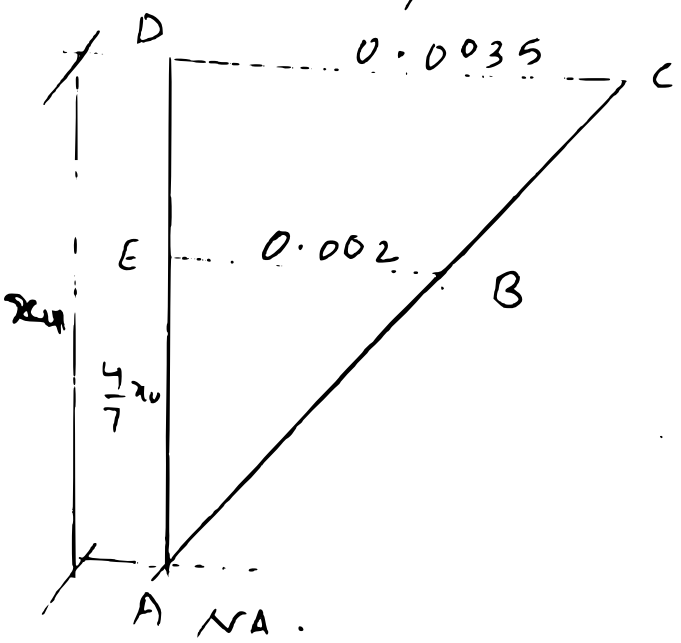
① Determination of tensile force.

Tinsile force = T = Tensile stress X Area of steel

$= 0.87 f_y \times A_{st}$.

② Determination of depth zones in concrete.

Diagrams for depth calculation.



using similar Δ rule

$$\frac{DC}{DA} = \frac{EB}{EA}$$

$$\frac{0.0035}{x_u} = \frac{0.002}{EA}$$

$$\Rightarrow EA = \frac{0.002\,x_u}{0.0035}$$

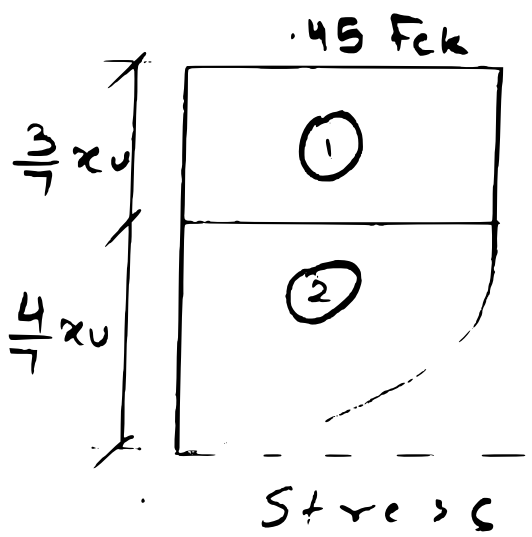
$$\Rightarrow EA = \frac{4}{7}x_u$$

$$AD = AE + ED$$

$$\Rightarrow \frac{4}{7}x_u + ED \Rightarrow ED = x_u - \frac{4}{7}x_u$$

$$ED = \frac{3}{7}x_u$$

Step 3 Calculation of compressive forces in concrete.



C_1 = Compressive force in part (1)

C_1 = stress x Area

$= 0.45 f_{ck} \times \left(\frac{3}{7}x_u \times B\right)$

$= 0.192\, B x_u f_{ck}$

C_2 = part (2)

= stress x Area

$$\frac{2}{3} \times \left[0.45 f_{ck} + 0\right] \times B \times \frac{4}{7}x_u$$

$$C_2 = \frac{2}{3} \times 0.45 f_{ck} \times \frac{4}{7}x_u \times B$$

$\Rightarrow C_2 = 0.171\ B x_u f_{ck}$

$C = C_1 + C_2 \cdots\cdots$

Total compressive force $C = 0.192\ B x_u f_{ck} + 0.171\ B x_u f_{ck}$

$$\boxed{C = 0.36\ f_{ck}\ B x_u}$$

13 march Class 2end

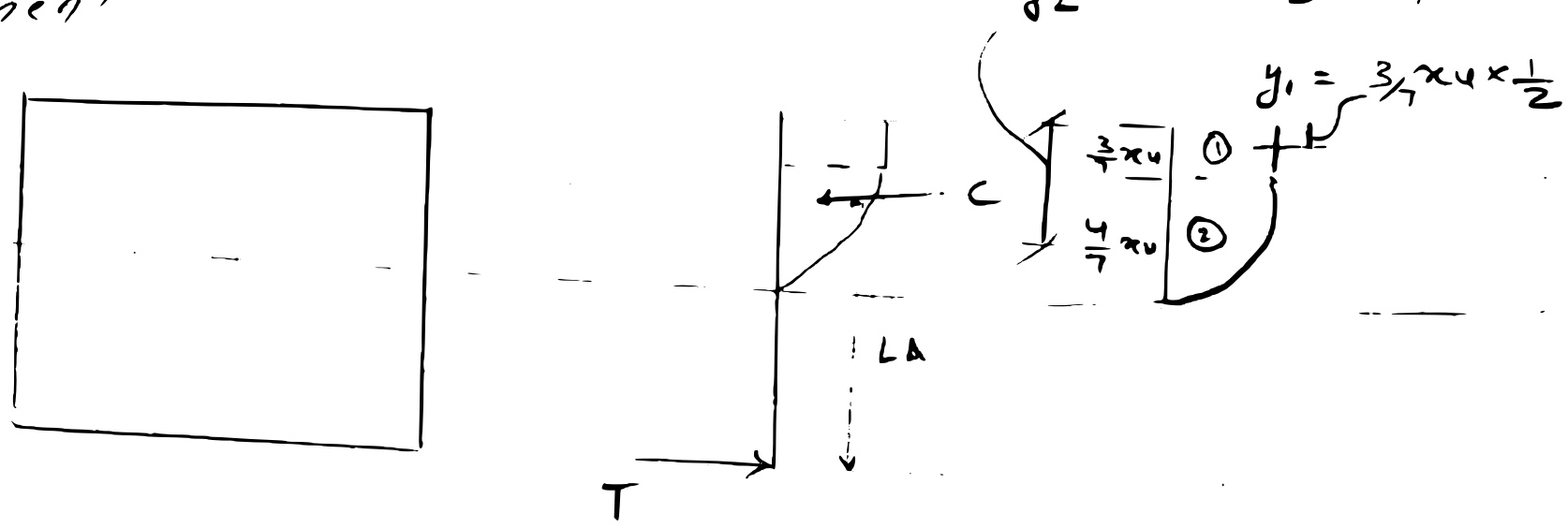
14 masch Class 3 start ↓

Step 4 Determination of lever arm

lever arm is the distance between compressive force and tensile force which causes moment in section.

$y_2 = \frac{3}{7} x_u + \frac{3}{8}^{th} \times \frac{4}{7} x_u$

$y_1 = \frac{3}{7} x_u \times \frac{1}{2}$



(h distance or distance of force "c" from top of fiber

$$\bar{y} = \frac{y_1 C_1 + y_2 C_2}{C_1 + C_2}$$

$y_1 = \frac{3}{7} x_u \times \frac{1}{2}$ (CG of rectangle)

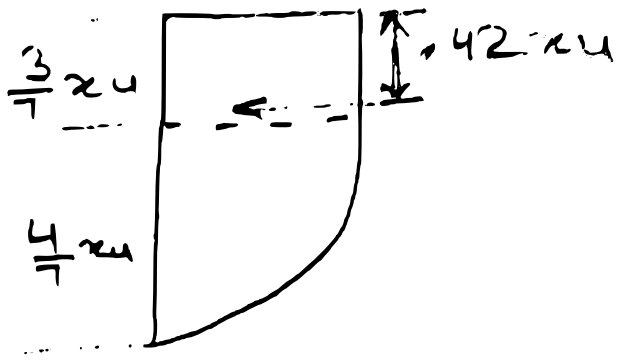
$C_1 = 0.192\ B x_u f_{ck}$.

$$y_2 = \frac{3}{7} x_u + \frac{3}{8} \times \frac{4}{7} x_u$$

$$C_2 = 0.171 B x_u f_{ck}$$

$$y = 0.416 x_u$$

$$\Rightarrow \boxed{\bar{y} \simeq .42 x_u}$$

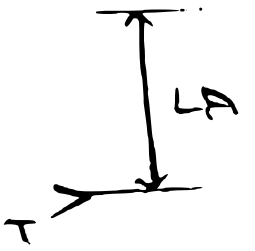


d is effective depth of beam
x_u is actual depth of NA.
B is width of section.

Steps Ultimate moment or ultimate moment carrying capacity

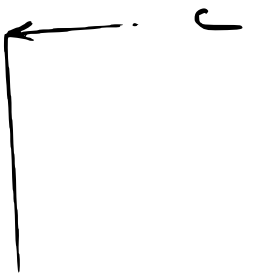
① From tension side

$$M_u = T \times L_A$$
$$= 0.87 A_{st} (d - 0.42 x_u)$$

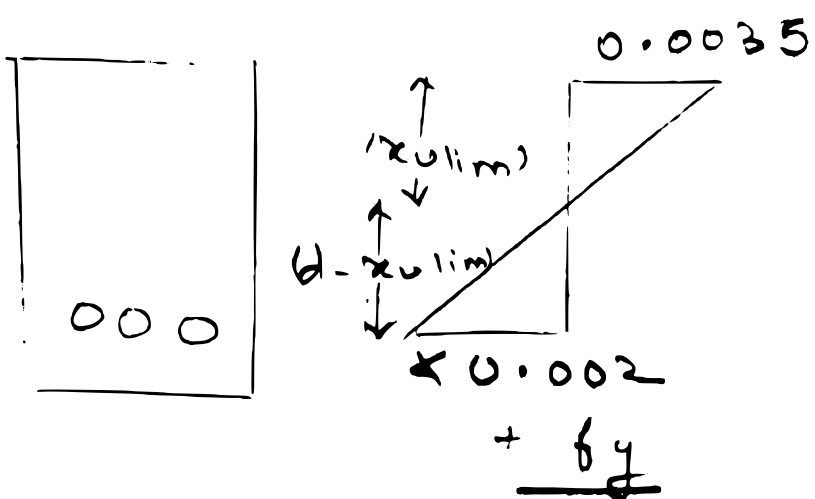


② from compression side.

$$M_u = C \times L_A$$
$$= 0.36 f_{ck} B x (d - 0.42 x_u)$$



Step 6 Determination of balanced depth or limiting depth of NA ($x_{u,lim}$)



Simmilar Δ rule

$$\frac{0.0035}{x_{ulim}} = \frac{0.002 + \frac{f_y}{1.15E_s}}{d - x_{ulim}}$$

$$x_{ulim} = \left(\frac{0.0035}{0.0055 + \frac{f_y}{1.15E_s}}\right) d \quad (1)$$

x_{ulim} or limiting depth of NA depends only on grade of steel (f_y)

Note: As the grade of steel increases x_{ulim} decreases.

$$x_{ulim} = \frac{0.0035\,d}{0.0055 + \frac{0.87 f_y}{2 \times 10^5}} \quad (2)$$

$$x_{ulim} = \frac{700\,d}{1100 + 0.87 f_y} \quad (3)$$

Grade of steel	$x_{u\,lim}$	$k = \frac{x_{u\,lim}}{d}$	ε
Fe - 250	0.53 d	0.53	0.00390
Fe - 415	0.48 d	0.48	0.00380
Fe 500	0.46 d	0.46	0.00417
Fe 550	0.44 d	0.44	0.0039

Given in IS code page 70 ↑

Step 7

Determination of actual depth of NA.

Actual depth of NA in limit state method can be determined by equating compressive force and tensile force.

$C = T \rightarrow 0.36 f_{ck} x_u b = 0.87 f_y A_{st}$

$$x_u = \frac{0.87 f_y A_{st}}{0.36 f_{ck} B}$$

Actual depth of NA depends on both grade of steel & grade of concrete f_y & f_{ck}

Step 8 Determination of area of steel or area of Rf.

$$A_{st} = 0.36 f_{ck} B x_u / 0.87 f_y$$

(9) Percentage of reinforcement reqd

$pt(\%)$

$$\frac{A_{st}}{B.d} \times 100\% = pt(\%)$$

$$\frac{0.36 f_{ck} B x_u}{Bd \times 0.87 f_y} \times 100\% = pt(\%)$$

$$\Rightarrow \frac{0.36 f_{ck} x_u}{0.87 f_y d} \times 100\% = pt(\%)$$

(10) Types of section or comparison of sections

1. When $x_u < x_{u\,lim}$.

if actual depth of NA is less than limiting depth of NA then it is called under reinforced section [URS]

in this section steel reaches yield value first while maximum strain in concrete is less than the failure strain. Hence this section indicates ductile failure & it is the most preferable section

M_u : T × LA

: $0.87 f_y A_{st} (d - 0.42 x_u)$

2. Balanced section

when $x_u = x_{u\,lim}$

When actual depth of NA is equal to limiting depth of NA then both steel & concrete reach to their ultimate strain simultaneously.

$M_{u\,lim}$: C × LA

: T × LA.

3. Over reinforced section

when $x_u > x_{u\,lim}$.

It is called over reinforced section In this section concrete reaches to its ultimate value first than steel. Brittle failure. Does not give warning.

$M_u = 0.36 f_{ck} x_u B (d - 0.42 x_u)$

Codal provisions.

Maximum % of Rf in tension or

maximum % of tension of

$P_t(\%)_{max}$ of 4 %. BD = Astmax

rgtr

or

$\frac{4}{100}$ BD

or

0.04 BD

Max area of compression of

$P_c(\%)_{max}$ of 4 %. BD = Asmax

or

$\frac{4}{100}$ BD

or

0.04 BD

Minimum area of reinforcement

(compression Rf or tension of)

$P_{min}\%$ of $\left(\frac{Astmin}{Bd} : \frac{0.85}{fy}\right)$

nless

Q. IES 2011: Modulus of elasticity of concrete is increased with - increase in age.

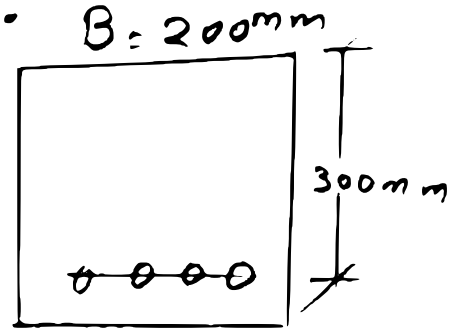
$$E_{co} \propto \frac{E}{1+\theta}$$

$$Age \propto \frac{1}{\theta}$$

$$E_{co} \propto \frac{1}{\theta}$$

$$\Rightarrow Age \propto E_{co}$$

Q Given a rectangular RL beam of effective cross section 200x300 grade of steel is Fe 415 and M25 grade concrete is used. if 4 numbers of 12 mm dia bars are used then find out ultimate moment or moment capacity for this section.



Fe 415 f_y = 415Mpa

M 25 f_{ck} = 25Mpa

4 bars @ 12 mm ϕ

M_u: ?

Ultimate term likhna matlab limit state method se solve karna padega.

SOL^n

Identify the section under reinforced

$x_{u\,lim}$: 0.48 d

Fe 415

d : 300 $\Rightarrow$ $x_{u\,lim}$: 0.48 × 300

= 144 mm.

Determination of x_u

$$x_u = \frac{0.87 f_y A_{st}}{0.36 f_{ck} B}$$

$A_{st} = n \times \frac{\pi}{4} \times \phi^2$

$A_{st} = 4 \times \frac{\pi}{4} \times 12^2$

$x_u = \frac{0.87 \times 415 \times 4 \times \frac{\pi}{4} \times 12^2}{0.36 \times 25 \times 200}$

$\Rightarrow x_u = 90.74$ mm

$x_u < x_{u\,lim}$

90.74 < 144 mm

URS

$M_u = T \times LA$

$= 0.87 f_y A_s + (d - 0.42 x_u)$

$= 0.87 f_y \times 415 \times 4 \times \frac{\pi}{4} \times 12^2$

$\times (300 - 0.42 \times 90.74)$

$= 42.77 \times 10^6$ Nmm or 42.77 kNm

for statically indeterminate beam the redistribution of moment allowed is 30%.

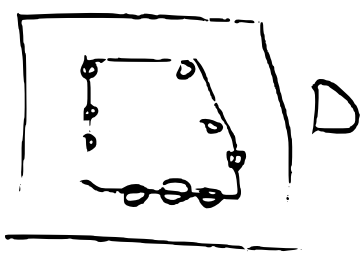
Expansion joint is provided when lateral dimension is more than or equal to 45 m.

Side face reinforcement

① when D_{total} depth > 750 mm.

A_{sf} = 0.1% of web area

= 0.1% BD



Side face on each side

$$= \frac{A_{sf}}{2} = \frac{0.1\%}{2} \text{ web area}$$

= 0.05% web area.

② The maximum spacing of side face of

not to 300 mm

or

Web thickness (B)

• ② When D > 750mm and the beam is subjected to torsion then side face reinforcement is provided.

Q In limit state approach spacing of main reinforcement controls primarily — cracking. !

Q. The maximum strain in concrete at the outermost compression fiber in the limit state design of flexural member is (as per IS 456 1978) = 0.0035

class 3 ends ↑

15 March 2024 class 4 start

Note: Minimum reinforcement is provided because ① to prevent sudden failure due to accidental increase in load

② To cater to secondary stresses such as temperature, corrosion etc.

Maximum reinforcement is provided to -

① to make comfort in placing of concrete

② to prevent difficulty in construction.

Maximum nominal size of aggregate taken is 20mm

Nominal maximum size of aggregate & adjustment

Size of mm	Adj to cem content kg/m³
	+40
10	0
20	-30.
40	

Grade of concrete. PCC RCC

① In general M-10 M-20

② Sea coast M-20 M-30

Pg 21 of code.

Note: In case of sea coast or sea water use of slag or pozzolana cement is advantageous.

While determining average or mean target strength formula $f_m = f_{avg} = f_{ck} + 1.65\sigma$

σ is standard deviation

	standard deviation
M10 to M15	3.5
M20 to M25	4
M30 to M60	5
M65 to M80	6

What is the value of mean strength for M20 grade concrete.

$f_m = f_{ck} + 1.65\sigma = 20 + 1.65 \times 4 = 26.6$ Mpa

Mix proportion as per IS 10262 code

frequency of sampling

Quantity of concrete in work m^3	No of samples
(1 to 5) till 5 m^3	1
(6 to 15) till 15 m^3	2
(16 to 30) till 30 m^3	3
(31 to 50) till 50	4
51 and above	4 plus one additional sample for 50 m^3 or part thereof

51 and above me 4 sample each 50 m^3 example 51 m^3 = plus 1 extra for 1

51 + 0 → 4 sample, 1 → 0. = 4 sample

100 m^3 = 51 + 49 → 4 cube, → 0 cube = 4 cube

101 m^3 : 5l + 50 m^2 : 5 cubes

↓ 4cubes ↓ 1cube

200 m^3 : 5l + 50 + 50 + 50 + 0 : 6 cubes

↓ 4 ↓ 1 ↓ 1 ↓ 0

The test results of sample shall be the average of strength of ~~three~~ 3 specimens, the individual variation should not be more than ±15% of average.

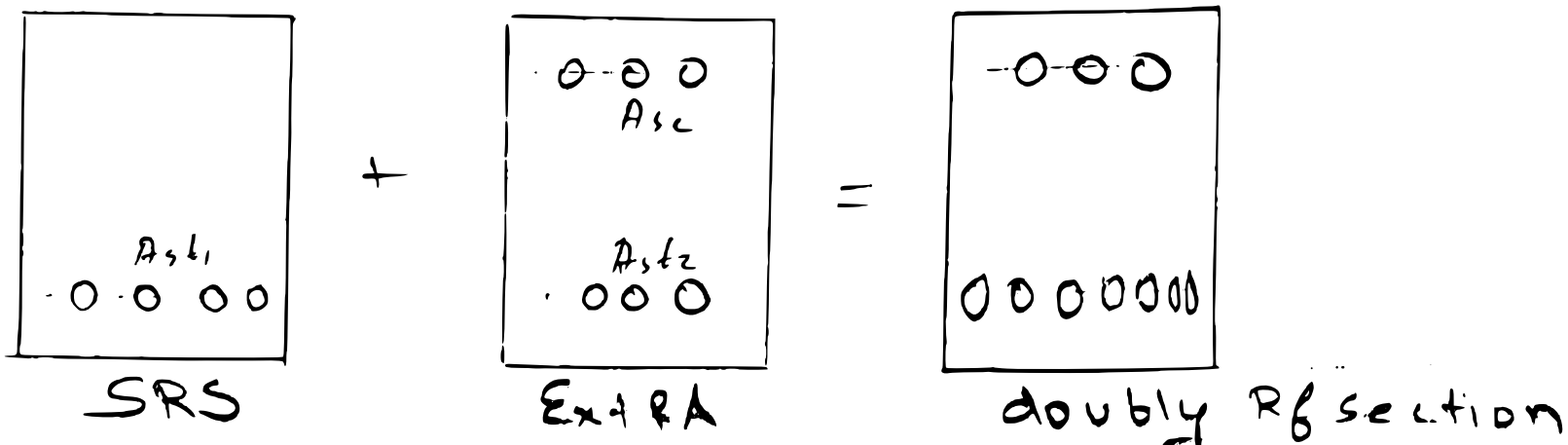
Singly reinforced beam or section:
When reinforcement is provided only on tension side then it is called singly reinforced section.

Doubly reinforced section or beam:
When reinforcement is provided in both tension & compression side then it is called doubly reinforced beam or section.

Necessity
1. When depth of beam is restricted
2. When width of beam is restricted
3. Reversal of loading
4. blasting load.
5. Impact load.

- Practically in all construction doubly reinforced section is used.
- In doubly reinforced section compression steel is under-stressed.

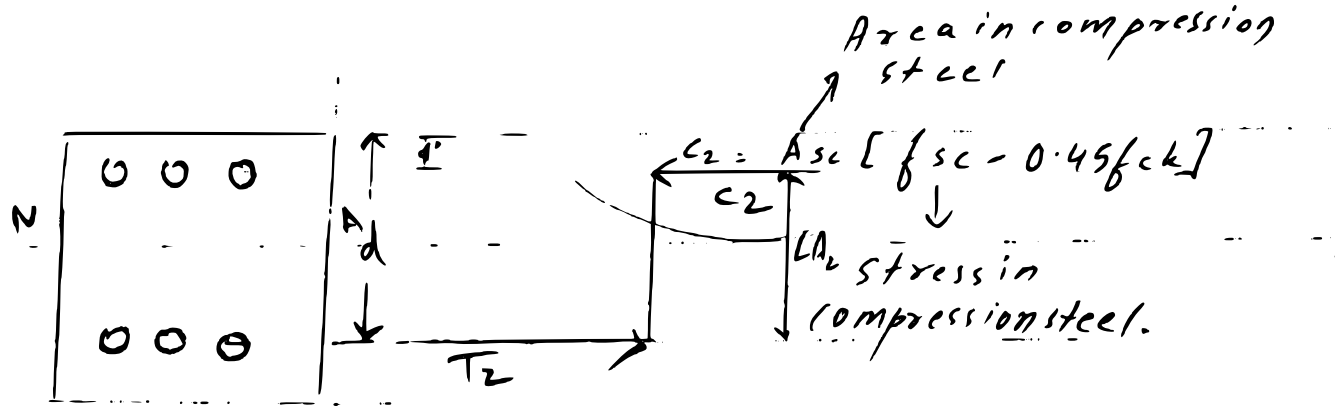
Design and analysis of doubly RC section



$C_1 = 0.36 f_{ck} x_u B$ $\quad C_2 = [f_{sc} - .45 f_{ck}] A_{sc}$ $\quad C = C_1 + C_2 = .36 f_{ck} B x_u + (f_{sc} - .45 f_{ck}) A_{sc}$

$T_1 = 0.87 f_y A_{st1}$ $\quad T_2 = 0.87 f_y A_{st2}$ $\quad T = T_1 + T_2 = .87 f_y A_{st1} + .87 f_y A_{st2} = .87 f_y [A_{st1} + A_{st2}]$

$LA_1 = (d - 0.42 x_u)$ $\quad LA_2 = (d - d')$



$T_2 = 0.87 f_y A_{st2}$

d' = distance of compression steel from top

Determination of ultimate moment.

(1) from compression side.

$M_u = C_1 LA_1 + C_2 LA_2$

$= 0.36 f_{ck} \; x_u b (d - 0.42 x_u) + (f_{sc} - 0.45 f_{ck})$ A_{sc} $(d - d')$

2) from tension side

$M_u = T_1 \times LA_1 + T_2 \times LA_2$

$= 0.87 f_y A_{st_1} (d - 0.42 x_u)$

$+$

$0.87 f_y A_{st_2} (d - d')$

Determination of area of steel

$M_u = M_{u1} + M_{u2}$.

$M_{u1} = M_{ulim}$ ↓ M_u lim.

$= T \times LA_2$

$= 0.87 f_y A_{st_1} (d - 0.42 x_u)$ ↓ x_u lim

$A_{st_1} = \dfrac{M_{ulim}}{0.87 f_y A_{st_1} (d - 0.42 x_u)}$

$BM = M_u = M_{u1} + M_{u2}$

↓

$BM = M_u = M_{ulim} + M_{u2}$

$\therefore M_u - M_{ulim} = M_{u2}$

$\Rightarrow BM - M_{ulim} = M_{u2}$

$M_{u2} = BM - M_{ulim} = T_2 \times LA_2$

$BM - M_{ulim} = 0.87 f_y A_{st_2} (d - d')$

$$\frac{BM - M_{ulim}}{0.87 f_y (d - d')} = A_{st2}$$

$$\Rightarrow M_{u2} = BM - M_{ulim} = C_2 \times LA_2$$

$$BM - M_{ulim} = (f_{sc} - 0.45 f_{ck}) A_{sc} (d - d')$$

$$\frac{(BM - M_{ulim})}{(f_{sc} - 0.45 f_{ck})(d - d')} = A_{sc}$$

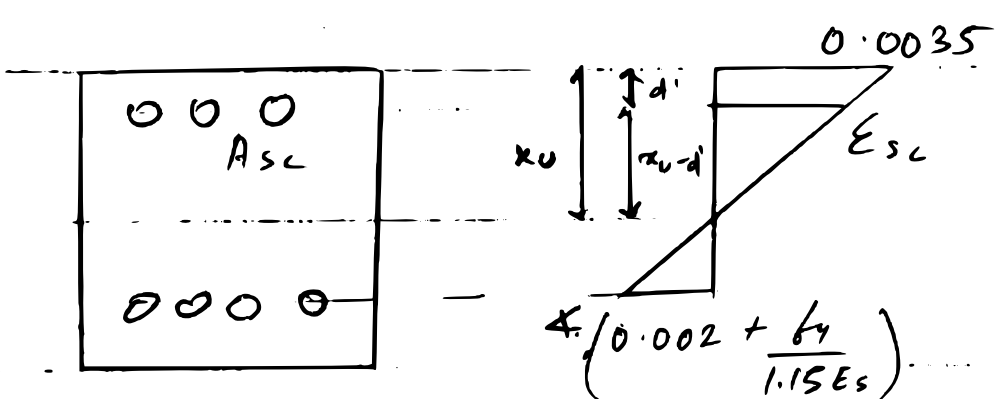
Shortcut formulae banane ka tareeka:
$M = F \times LA = \sigma \times A \times LA \Rightarrow \frac{M}{\sigma \times LA} = A$

$$A_{st1} = \frac{M_{ulim}}{0.87 f_y (d - 0.42 x_{ulim})}$$

$$A_{st2} = \frac{M_{u2} \rightarrow (M_u - M_{ulim})}{0.87 f_y (d - d')}$$

$$A_{sc} = \frac{M_{u2} \rightarrow (BM - M_{ulim})}{(f_{sc} - 0.45 f_{ck})(d - d')}$$

Total tensile steel $A_{st} = (A_{st1} + A_{st2})$.



ε_{sc} is strain in compression steel

$$\frac{0.0035}{x_u} = \frac{\varepsilon_{sc}}{x_u - d'} \Rightarrow \varepsilon_{sc} = 0.0035 \frac{(x_u - d')}{x_u}$$

$$0.0035\left[\frac{x_u}{x_u} - \frac{d'}{x_u}\right] = \varepsilon_{sc}$$

$$\boxed{0.0035\left[1 - \frac{d'}{x_u}\right] = \varepsilon_{sc}}$$

x_u is actual depth of NA

ε_{sc} is strain @ in compression steel

. In absence of any data, maximum strain in concrete / total shrinkage strain is taken as .... 0.0003 or 3×10^{-4}

Code of prestress IS 1343

Q What is the short term modulus of elasticity of concrete for M25 grade & also, state its range.

Ans $E_c = 5000\sqrt{f_{ck}}$

$= 5000\sqrt{25}$

$= 25,000\ N/mm^2 \pm 20\%$

$E_c \times 20\% = 25000 \times \frac{20}{100} = 5000$

Range : $E_{c-20\%} = 25,000 - 5000 = 20,000$

$E_{c+20\%} = 25,000 + 5000 = 30,000$

$\Rightarrow$ Range = 20,000 to 30,000.

M30 k liye banana h same gn Ko.

Class 4 ends

16 March 2024 Class 5 start.

T beam & L beam (end beam)

A slab which is assumed to act as a compression flange of T beam or L beam shall satisfy the following conditions:

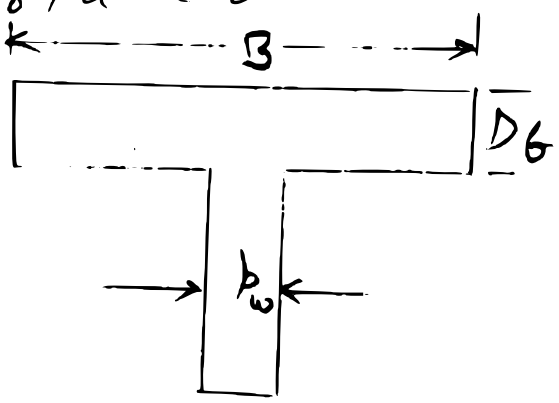
1) The slab shall cast integrally with web and the slab effectively bonded together in any manner

2) if the main reinforcement of the slab is parallel to beam, transverse reinforcement shall be provided perpendicular to it.

Effective width of flange as per IS 456.

In absence of any data, the following formula is used to find out effective width

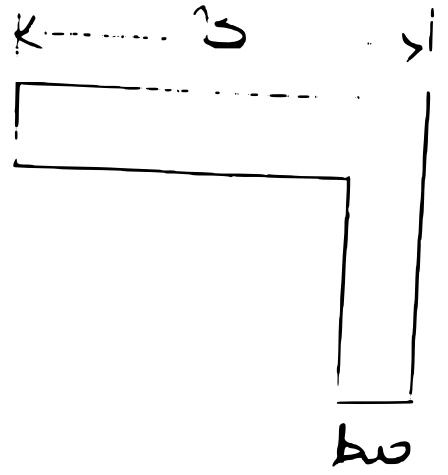
1) Isolated T beam.



$$B_f = \frac{l_o}{\frac{l_o}{B} + 4} + b_w$$

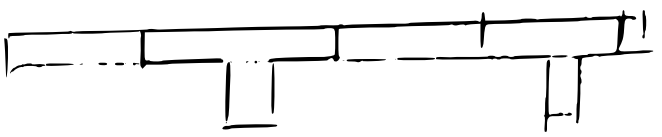
effective width of flange $= \frac{\text{unsuppl length of flange}}{\frac{\text{unsuppl}}{\text{total width}} + 4}$ + width of web.

2) Isolated L beam.



$$B_f = \frac{1}{2}\left[\frac{l_0}{\frac{l_0}{6}+4}\right] + b_w$$

3) Monolithic or continuous beam



a) T beam.

$$B_f = \frac{l_0}{6} + 6D_f + b_w$$

= centre to centre spacing between 2 beams

= $b_w + \left(\frac{l_1}{2}\right) + \frac{l_2}{2}$.

↓ less

b) L beam

$$B_f = \frac{l_0}{10} \times \frac{1}{2} + 6D_f \times \frac{1}{2} + b_w$$

or $\frac{l_0}{12} + 3D_f + b_w$

= $b_w + \frac{l_1}{2}$

↓ less

l_0 is distance b/n points of zero moment in the beam

l_i is clear ..... spacing between adjacent beams.

if beam is simply supported then consider l_o = effective span.

for continuous beam and frame assume l_o : 0.7 times of effective span.

Codal provisions. code Pg 22

1) limits of chloride content of concrete

types or uses of concrete	Maximum total acid soluble chloride content expressed as kg/m³ of concrete
a) concrete containing metal & steamcured at elevated temperature and prestressed conc.	0.4
b) Reinforced concrete or plain concrete containing embedded metal	0.6
c) concrete not containing any embedded metal or requiring protection from chloride	3

Use of low alkali OPC having total alkali content not more than 0.6 percent (as Na_2O equivalent). 5.4.2.6

Limit State of collapse - Shear

Shear is the second important design criteria for beam or bending member whereas bending or bending moments is primary design criteria.

Shear resistance is obtained by —

a) Shear resistance without shear reinforcement.
   - If shear reinforcement is not provided then also beam has some shear resistance —
     - main tension reinforcement (dowel action of main bars)
     (Pt %.
     - Aggregate interlocking
     - uncracked concrete ie grade of concrete.

∴ Shear resistance τ_c without shear reinforcement : $\tau_c = (1) + (2) + (3)$

τ_c is design shear strength of concrete

Note : if no shear reinforcement is provided then the crack angle is 30°. with horizontal for shear crack.

b) Shear resistance with shear of

- provided in form of stirrups or transverse reinforcement.

1) Vertical stirrups
2) Bent up bars
3) Inclined stirrups

- Practically preferable is vertical stirrup but best stirrup to take shear is inclined stirrups.
- In shear reinforcement, important criteria is spacing.
- Maximum spacing between stirrups
  - Vertical stirrups
    - max spacing $\not\propto .75d$, $\not\propto 300mm$, $\not\propto S_v$

d is effective depth of beam

$$S_v = \frac{0.87 f_y A_{sv} d}{V_{us}}$$

f_y = char str of stirrups

A_{sv} = Area of shear stirrups

V_{us} = external design shear or Shear force which is to be carried by shear reinforcement for which shear is designed

Maximum spacing for inclined stirrups: 5.6

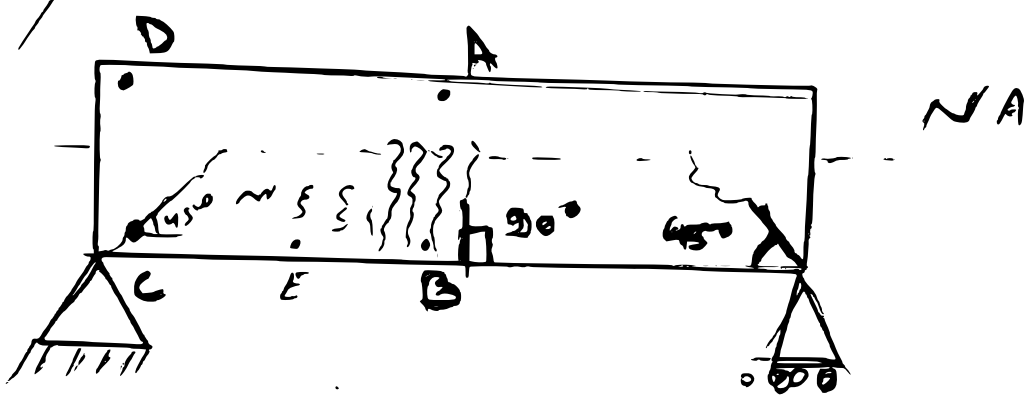
max spacing $\begin{cases} \not\propto d \\ \not\propto 300mm \\ \not\propto S_v \end{cases}$ ↓

$$S_v = \frac{0.87 f_y A_{sv} d}{V_{us}} (\sin\alpha + \cos\alpha)$$

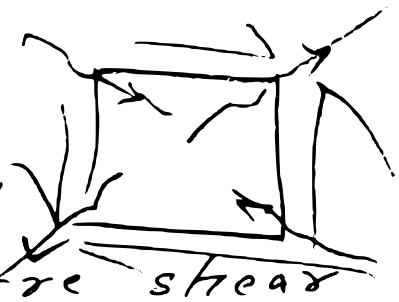
α is angle of inclination of inclined bar with horizontal. (45° in general)

3) Bent up bars – bent up bars should be bent @ lever arm distance.

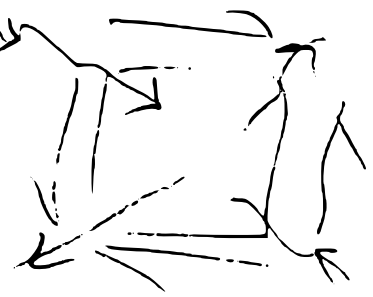
Crack patterns:



Point C: diagonal tension crack.
- No bending moment
- Only shear force present hence pure shear condition and shear crack occurs



Point D: Diagonal compression crack
- Critical failure case (condn
- (cause) sudden failure
- (concrete gets crushed like powder
- Most dangerous failure!



Point B: flexure or bending or rupture crack
- No shear force, only bending moment.

Hence vertical crack develops at 90° with horizontal.

Point C flexure shear crack.
Both bending moment and shear force present.
crack angle more than 45° & less than 90°.
$45° < \theta < 90°$

Design.
Step 1 Shear resistance or design shear: τ_c.

$$\tau_c = \frac{V_c}{bd}$$

τ_c value is directly given in code which is based on grade of concrete (M) and percentage of tension reinforcement (p_t %).

Step 2 τ_v nominal external shear stress (due to load)

$$\tau_v = \frac{V_u}{bd}$$

V_u is shear force due to design load

τ_{vs} : Design shear strength for shear reinforcement or shear reinforcement shall be designed to carry τ_{vs} shear stress.

$$\tau_{vs} = \tau_v - \tau_c$$

$$\tau_{vs} : V_{us} : \frac{V_u}{bd} - \frac{V_c}{bd}$$

$V_{vs} = V_{us}$: ... $- \tau_{us} \times$ Area

vertical stirrups @ $V_{us} = \frac{0.87 f_y A_{sv} d}{S_v}$

Inclined stirrup @ $V_{us} = \frac{0.87 f_y A_{sv} d}{S_v} (\sin\alpha + \cos\alpha)$

Maximum shear stress $\tau_{c\,max}$ Value given in IS code based on grade of concrete only.

Q Numerical questions

Q1 Ko banaya h sheet 1 me.

Q2 ka ans c Q3 ka ⑤

Q4 b. Q5 c Q6 b Q7 skip Q8 skip Q9 c

Q10 d. 11 c 12 b 13 c 14 a

Class End

20 March 2024 class 6 start

Analysis & design of shear strength of concrete

τ_c = design shear ..... of concrete

τ_v = nominal shear stress (due to load)

τ_{cmax} = maximum ..... shear stress of concrete

Conditions

(1) When $\tau_v < \tau_c/2$ (Capacity load hese bhi kam load aaya)
or $\tau_v < 0.5\tau_c$

- Very safe in shear hence no shear reinforcement is required to design and no need of minimum shear reinforcement

also:

- no need of shear design
- no need of minimum shear reinforcement

(2) When $\frac{\tau_c}{2} < \tau_v < \tau_c$ or $\tau_v < \tau_c$

. This case is also safe in shear hence no need to design shear reinforcement but minimum shear reinforcement is provided in form of vertical stirrups.

(3) When $\tau_c < \tau_v$

for this case shear reinforcement is necessarily required to design.

which can be provided in form of vertical stirrups or bent up bars or inclined stirrups.

Case IV $\tau_{c,max} < \tau_v$ (analogy paper electrical leg gasaj)

$\tau_{c\,max} < \tau_v$ is unsafe condition redesign the section. by -

- increasing the crossectional area so that τ_v decreases.

In condition (1) & (3) the checkings are always less than τ_{cmax}.

Design:

1. Minimum shear reinforcement:

formula $\frac{A_{sv\,min}}{b s_v} = \frac{0.4}{0.87 f_y}$ (Limit state design meth.)

or $A_{sv\,min} = \left(\frac{0.4}{0.87 f_y}\right) b * s_v$

b is width of beam

S_v is spacing of the ... stirrups

Design shear reinforcement :-

Design shear reinforcement can be provided in any of the following forms -

1. Vertical stirrups —

$$V_{us} = \left(\frac{0.87 f_y A_{sv} d}{S_v} \right)$$

2. Inclined stirrups —

$$V_{us} = \left(\frac{0.87 f_y A_{sv} d}{S_v} \right) (\sin\alpha + \cos\alpha)$$

3. Bent up bars —

- Bent up bars are always provided with vertical stirrups.
- Bent up bars are provided by bending main reinforcement.
- Bending may cause stress concentration at bend portion.
- When bent up bar is used, 50% shear is taken by bent up bar & remaining 50% is achieved by stirrups.

τ_{cmax} table

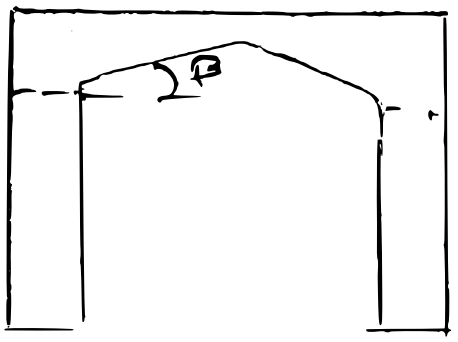
concgrade	M15	20	25	30	35	40
τ_{cmax}	2.5 N/mm^2	2.8	3.1	3.5	3.7	4.0

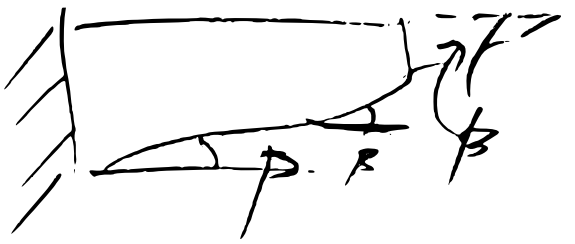
As the steel grade increases shear strength also increases but ductility decreases

Note: for shear reinforcement maximum grade of steel provided is Fe 415.

Fe 250 grade sakta, 415 de sakta but isse upar grade of steel provide nahi kar sakte kyonki ductility decrease hone ke kaaran cracks develop hone lagte hai.

Tapered Section or beam of varying depths.





Nominal Shear stress

$$\tau_v = \left[\frac{V_u \pm \frac{M_u}{d}\tan\beta}{bd}\right]$$

Nominal shear stress for beam with uniform depth. $\tau_v = \frac{V_u}{bd}$

β is angle between top & bottom edge

M_u is bending moment at the section.

Mu and Vu are bending moment and shear force at the given section.

Minus sign or negative sign is adopted when bending moment Mu decreases - increases as well as effective depth increases ⊖ Mu ↑ d ↑

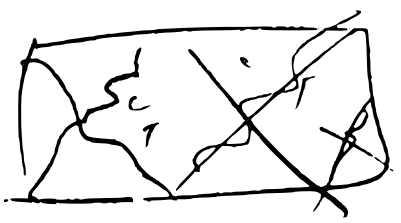
⊕ sign is adopted when Mu decreases as depth increases ⊕ Mu ↓ d ↑

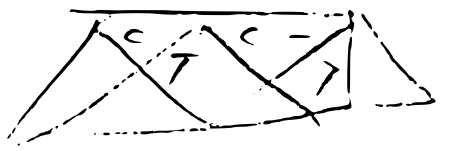
d is effective depth at the section

Note: as the grade of concrete increase shear strength capacity increases

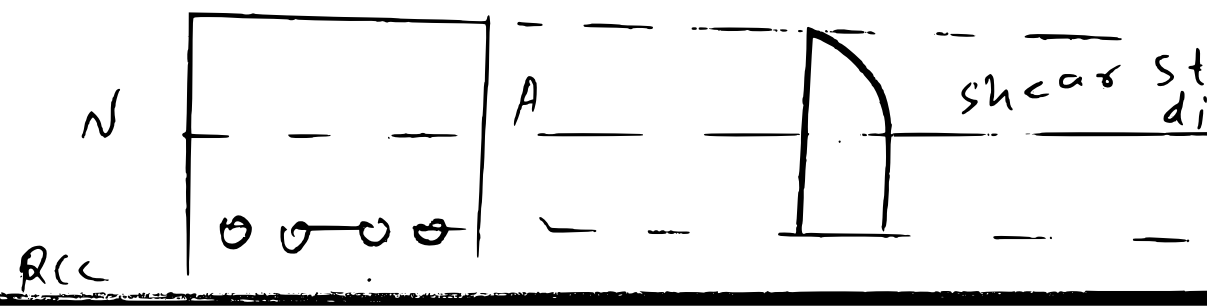
As the grade of concrete increases brittleness decreases

Shear stress theory can be compared with 2D truss analogy = Plane truss analogy or lattice girder analogy / lattice girder theory.





Shear stress distribution in RCC.



Shear stress variation in RCC — parabolic upto neutral axis from top fiber and then constant beyond it upto steel level / tension reinforcement level.

Nominal cover to ~~ned~~ tey meet durability requirements Pg 47

Exposure	Nominal cover
mild	20
moderate	30
Severe	45
Very severe	50
Extreme	75

Subclause main bar diameter upto 12mm (6mm 8mm 10mm 12mm) is used matlab structure utna important nahi hai islye cover 20 mm ma de kar 20-5 matlab 15mm denge mild exposure

Subclause d, 40mm & 45mm

Cover value Should not be less than these values given in table It can be increased by +10mm and can be decreased by -0mm

Subclause d: if severe & very severe exposure condition and M35 & above grade concrete is used matlab hume high grade ka concrete use kiya hai isilye cover 5mm reduce kr skte hain

Numerical questions: Worksheet number 3

Q: In RCC beams as the i/ge of Ast increases, depth of NA → depth of NA increases.

$0.87 f_y A_{st} = 0.36 f_{ck} x_u \Rightarrow A_{st} \propto x_u$

Q3. A floor slab of thickness t is cast monolithically transverse to rectangular continuous beam of span l & width B. if the distance between two consecutive points of contraflexure is l_o. the effective width of compression flange at continuous support is →

$B + 6t + l_o/6$ $\quad B_f = \frac{l_o}{6} + (l)_f + b_w$.

Q4 The side face of if reqd in beam is
→ 0.1% of web area

Q5 Minimum clear cover to the main steel bars in slab beam columns & footing respectively are=
→ as per IS 2540 75

Q7. if the depth of actual NA is more than the beam is depth of critical neutral axis → Over reinforced sec.

Q8 What is the adoptable maximum spacing between vertical stirrups in RCC beam of rect cross section having effective depth 300 mm.

Iss Rn me Sv ko check nahi kiya
Kyonk Sv ka liye fy wagar naidiya

minimum shear reinforcement
vertical stirrups ke form me
dete hain.

15	20	25	30	35	40
2.5	2.8	3.1	3.5	3.7	4.0

V.7s of } .75/300, 225 } ⇒ Ans. 225mm
∅ 8v
∅ 3mmm } 300

Q12 for an RCC beam the shape of shear stress diagram is → ⓒ parabolic above NA & rect below NA

Q13 if nominal shear stress does not exceed the permissible shear stress → minimum shear reinforcement is still provided. $\tau_v < \tau_c$.

Q14. for M20 grade of concrete maximum shear stress shall not exceed → 2.8 Mpa.

Q15 How can shear strength be enhanced on a beam → By providing stirrups.

Q16 In a reinforced concrete wall shear key is provided if → when retaining wall is not safe against sliding.

basic end

21 March 2024 Class 7 start

Bond stress is shear stress between concrete and steel.

It is obtained by strain compatability

Hence it is based on maximum principal strain theory St. venanti theory.

Bond is holding action between concrete and steel.

Bond is obtained by or due to.

(1) Adhesion: sticking action between two different materials.

(2) Cohesion: Sticking action between same material.

(3) frictional resistance: friction of concrete due to aggregate interlocking.

(4) Mechanical resistance: Grip due to twisted surface of steel

Note: Mechanical resistance is not obtaind in MS or plain bars

Q 1999 IES: Bond is *longitudnal* shear stress

Types of bond provided as per IS code.

1) end anchorage bond: It is critical & beam may fail due to lack of this bond.

2) flexural bond: This bond is required at the mid span in flexure zone in order to maintain flexural compatability

Bond stress:

$$\tau_{bd} = \frac{\text{Shear force}}{\text{Bond resistance area}}$$

Bond resistance area $= \frac{\text{perimetral area}}{n \times (\pi \times \phi)}$

$$\tau_{bd} = \frac{V}{n\pi\phi \times jd}$$

n is number of bars

ϕ is diameter of bar.

$$\tau_{bd} = \frac{V}{\leq 0 \; jd.}$$

$\downarrow$

$n\pi\phi$

jd is lever arm distance

τ_{bd} is design bond stress.

code page 93. IS:456.

Design bond stress τ_{bd} in limit state method for plain bars in tension shall be as follows

Grade of Concrete	M20	M25	M30	M35	M40
τ_{bd} N/mm²	1.2	1.4	1.5	1.7	1.9

Note: if deformed bar or HYSD bar is used then τ_{bd} value should be increased by 60% (or multiplying factor (1.60))

Note: if bars are in compression then τ_{bd} value should be increased by 25%. (multiplying Factor 1.25)

Bars bundled in contact.

Bundling of bars is done when desirable diameter is not available

When bundling is done, developement length of each bar shall be increased by -

when 2 bars bundled = l_d ↑ by 10%.

3 " " : " ↑ " 20%.

4 " " : " " " 33.0%.

Permissible bond stress for plain bars in tension. M20 → 0.8 N/mm² (WSM code)
M25 → 0.9 N/mm²

Anchoring reinforcing bar

1) Anchoring bars in tension:
hooks and bends are provided
Bend: The anchorange value of bend shall be taken as 4 times the diameter of bar for each 45° degree bend. max upto 16 times.

Hook: The anchorage value of standard U hook shall be equal to 16 times the diameter of bar

Hooks shall normally be provided for plane bars in tension - Mild steel only.

	development length l_d
45°	4 ϕ
90°	8 ϕ
135°	12 ϕ
180°	16 ϕ

ϕ is diameter of bar

Developement length Ld.

In order to maintain the position of bar to satisfy bond comptability bond force should be equal to tensile force applied externally.

safety in bond Bond force // Tensile force.

$\tau_{bd} \times n\pi\phi L_d = 0.87 f_y \times n \cdot \frac{\pi}{4}\phi^2$

$L_d = \frac{0.87 f_y \phi}{4 \tau_{bd}}$ (LSM)

(WSM) $L_d = \frac{\sigma_{st}\phi}{4\tau_{bd}}$

σ_{st} is permissible tensile stress in steel

lsm developement length = 0.87 x characteristic stress of steel x diameter of bar / 4 x design bond strength.

for Fe 250 or mild steel σ_{st} = 140 Mpa

for Hysd bar 415 σ_{st} = 230 Mpa or 230 N/mm²

for column permissible stresses in compression steel σ_{sc} = 150 Mpa for MS

190 Mpa for HYSD

Tbd value as per Is code table depends on grade of concrete only.

Codal provision check for developement length for flexural bond.

formula: 1) for continuous or fixed support

$$Ld \leq \left(\frac{M_1}{V} + L_0\right)$$

2) for simply supported end.

$$Ld \leq \left(1.3\,\frac{M_1}{V} + L_0\right)$$

Notations

developement length $\leq 1.3 \times \frac{\text{Moment of resistance of section}}{\text{shear force at the section.}}$ x summation of anchorage beyond the centere of support & the equivalent anchorage value of any hook or bend

for simply supported beam $\frac{M_1}{V}$ value shall be increased by 30%. ~~extra~~

L_0 should not be less than

L_0 $\left\{\begin{array}{l} \text{effective depth} \\ \text{of beam or} \\ 12\phi \end{array}\right\}$ whichever is more.

Note: to check bond lab test done: Pull out test.

Splicing of bars or splicing of reinforcement

lapsplicing:

lap splicing is done for bar diameter upto 36 mm.

Bar diameter more than 36 mm can be welded only.

lap length

1) Direct tension.

minimum lap length $\geq 24d$

$l_{p\,minimum} \geq 300$

: agar do different diameter ke bars honge to minimum diameter of bar or thinner bar ka diameter use karna hai

2) Bending Tension.

minimum lap length $\geq ld$

≥ 300

3) Direct compression

$l_{p,min} \geq Ld.$

≥ 240

When bond criteria in beam does not get satisfied in beam in that case provide more number of bars of lesser diameter or thinner diameter.

Note: thinner bars use karne se Specific surface area badh jayega jisse grip ya bond accha ho jayega.

Q1 The average permissible stress in bond for plain bars in tension
→ increased by 25% for bars in compression

Q2: The main reason of providing no of reinforcing bars at suppor in SSB to resist in that zone
→ Shear stress & bond stress. (bc both)

Q3: The development length in compression for 20 mm dia deformed bar of grade fe 415 embedded in concrete M25 whose design bond stress (τ_{bd}) 1.40 is.

$$L_d = \frac{0.87 f_y \phi}{4 \tau_{bd}}$$

$$\tau_{bd} = 1.4 \times 1.25 \times 1.60$$

$$L_d = \frac{0.87 \times 415 \times 20}{4 \times 2.8} \quad : \underset{(d)}{644.7 \text{ mm.}}$$

Q4 Which of the following is correct expression to estimate the ld of deformed bas as per IS456:2000 in LS design.

Ans $\frac{\sigma_s \phi}{6.4 \tau_{bd}}$ or $\frac{0.87 f_y \phi}{4 \times (1.6 \times \tau_{bd})}$

Q5 What is is the bond stress acting parallel to reinforcing bar on interface between bar and concrete → shear stress.

Q7 Consider true or false statement.

1) Modulus of elasticity of concrete increases with increase in compressive strength of concrete

Ans True

$E_c = 5000 \sqrt{f_{ck}}$

2) Brittleness of concrete increases with decrease in compressive strength of conc.

Ans True

7.10

3) Shear strength of concrete increase with increase in compressive strength of concrete

True.
- Zcmax kg table

→ All 3 are correct statements.
Class Tends

22 March 2024 Class 8 (World water day) (newchap)

Limit state of Torsion or twisting

Torsion or twisting moment acts about longitudnal axis of the member. Hence it affects the total depth of cross section.

Torsion causes shear stress

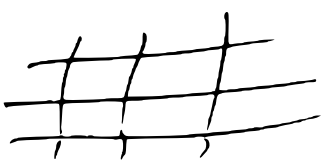
Types of torsion:

1) Primary or equilibrium torsion or direct or determinate torsion

Example: end beam, (Lbeam) deep beam, ring beam, (underwatertank) canopy or portico beam.

2) Secondary or indirect or indetermin-ate or compatability torsion.

Example: Grid floor beam.



Design and analysis of torsion

1. Shear force due to torsion

$$V_t = 1.6\left(\frac{T_u}{b}\right)$$

T_u : torsional moment

b : width of beam

2) Bending moment / Moment due to torsion #.

$$M_t = \frac{T_u}{1.7}\left[1 + \frac{D}{b}\right]$$

D : total depth of beam.

3) Equivalent shear force.

$$V_{ue} = V_u + V_t$$

$$= V_u + 1.6\left(\frac{T_u}{b}\right)$$

V_u is external shear force due to loading.

4) Equivalent moment.

$$M_{ue} = M_u + M_t$$

$$M_{ue} = M_u + \left[\frac{T_u}{1.7}\left(1 + \frac{D}{b}\right)\right]$$

(a) if $M_u > M_t$

$M_{ue} = M_u$.

means no need to provide compression reinforcement.

(b) if $M_u < M_t$

$M_{ue} = (M_u + M_t)$

provide compression reinforcement for torsion. Compression reinforcement is provided in form of longitudinal reinforcement in flexural compression flange.

(beams in torsion

(i) $\tau_{ve} < \tau_c/2$: Very safe in shear as well as in torsion.

No need of minimum shear reinforcement as well as no need of minimum torsional shear reinforcement.

(ii) $\frac{\tau_c}{2} < \tau_{ve} < \tau_c$ or $\tau_{ve} < \tau_c$

minimum shear reinforcement is provided [in form of vertical stirrups] but no need of minimum torsional shear reinforcement.

(iii) $\tau_c < \tau_{ve}$: shear reinforcement is required to be designed as well as torsional shear reinforcement is required to be designed.

Note: Torsional reinforcement is provided in form of longitudinal reinforcement whereas shear reinforcement is called transverse reinforcement.

(iv) $\tau_{c,max} > \tau_{ve}$. unsafe hence in this case redesign the section.

τ_{ve} = equivalent nominal shear stress.

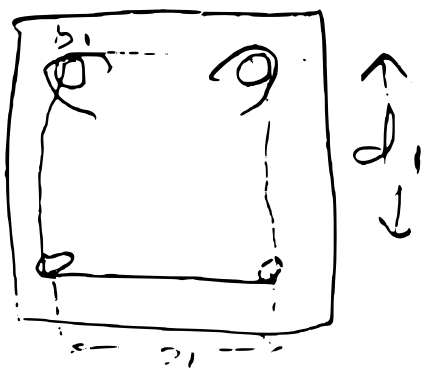
$$\tau_{ve} = \left(\frac{V_{ve}}{bd}\right) \quad \{\text{code pg } 75\}$$

Transverse reinforcement area

$$\text{Area } A_{sv}\left[\frac{V_u}{2.5} * \frac{1}{0.87 f_y} * \left(\frac{S_v}{d_1}\right)\right]$$

$$+$$

$$\left[\left(\frac{T_u}{b_1}\right)\left(\frac{1}{0.87 f_y}\right)\left(\frac{S_v}{d_1}\right)\right]$$



T_u is torsional moment

V_u is shear force (due to external loading)

b_1 is centre to centre distance between corner bars in the direction of width.

d_1 is centre to centre distance between corner bars along the depth.

f_y is characteristic strength of reinforcement

τ_c is design shear strength of concrete

Spacing : Maximum spacing of torsional shear reinforcement

- $\frac{x_1 + y_1}{4}$
- x_1
- 300 mm

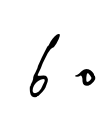
} ↓ less.

if total depth of beam is more than 450mm and torsion acts in that case side face reinforcement is provided

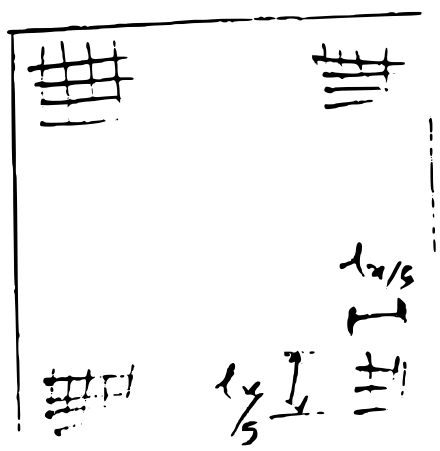
A_{sf} = 0.1 % web area. (both sides)

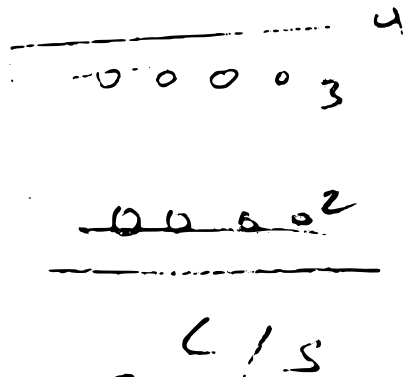
A_{sf} (each face) : $\frac{0.1\%}{2}$ web area ... = 0.05% of web area

Maximum spacing of side face reinforcement should not be more than 300 mm or width of section whichever is less.



Note: torsion affects both top fiber and bottom fibers ie complete depth of member hence torsional reinforcement should be provided @ top & bottom fibers & if it generally provided in 4 layers. with length of each layer $\frac{l_x}{5}$. where l_x is shorter span





Torsion theory is based on 3D truss analogy or space truss analogy or skew bending theory (Mc Ue's theory)

23 March 2024 Class 9

Rcc worksheet 4 Qn 10 explanation.

beam P:

$\tau_c = 0.75\ N/mm^2$

$\tau_v = \frac{V_u}{bd} = 1.33\ N/mm^2$

$V_{us} = V_u - V_c$

$= 400\ kN - [\tau_c \times bd]$

$= [400 \times 10^3\ N - 0.75 \times 400 \times 750]$

$V_{us} = 175 \times 10^3\ N$

Rcc worksheet 4,98 explanation.

Q8. $\tau_{ve} < \tau_c$ given.

$V_u = 20\ kN$

$M_u = 200\ kN-m$

$T_u = 9\ kNm$

$b = 300\ mm$

$D = 425\ mm$

$d_c = 25\ mm$

(A) $V_{ve} = ?$

$V_{ve} = V_u + 1.6 \frac{T_u}{b}$

$= 20 + 1.6 \times \frac{9}{0.3} = 68\ kN$

(B) $M_{ve} = ?$

$M_{ve} = M_u + \left(\frac{T_u}{1.7}\left(1 + \frac{D}{b}\right)\right)$ ignore.

$M_{ve} = M_u = 200\ kNm.$

Question me $\tau_{ve} < \tau_c$ diya hai it means minimum shear reinforcement should be provided but no need of torsional reinforcement